
Supplementary Information

Beyond Geometric Preorganization: A Bottleneck-Guided Pre-Reaction-State Framework for PET Hydrolase Engineering

Detailed methods and supporting analyses for FAST-PETase engineering

Contents

Supplementary Methods

- S1. Experimental PET-film hydrolysis assays
- S2. FAST-PETase system preparation and PET model
- S3. Docking and molecular-dynamics simulations
- S4. Replica design, PRS sampling and statistical analysis
- S5. Quantum-chemical and electronic-structure analysis
- S6. Distortion/interaction activation-strain analysis

Supplementary Results

- S7. Additional characterization of Y87Q
- S8. Additional characterization of S187Y and cross-scaffold validation
- S9. Ensemble quantum-chemical analysis and mechanistic interpretation

Supplementary Discussion

- S10. Scope and limitations of the PRS quantity-quality framework

Supplementary References

Supplementary Figure 1

Supplementary Methods

S1. Experimental PET-film hydrolysis assays

The enzyme preparation was tested with amorphous PET film as the substrate. For the temperature gradient test of Y87Q mutant, the purified enzyme was reacted with glycol-PET films at 40-70 °C for 120 h before the determination of the soluble hydrolysis products TPA, MHET, and BHET. Reaction time course experiments were conducted at 60 °C while pH activity was determined within 48 h at pH 5.5-8.5. For the active site mutants screen, the purified enzyme was incubated with the amorphous PET film at 40 °C for 72 h. The soluble product was analyzed by 260 nm absorbance while TPA and MHET were quantified via UPLC. The reaction system in Figure 3 workflow consists of 50 mM glycine-NaOH (pH 9), 50 mM NaCl, 10% (v/v) DMSO, and 150 nM purified protein.

S2. FAST-PETase system preparation and PET model

Three enzyme-substrate systems were considered in parallel: FAST-PETase, FAST-Y87Q and FAST-S187Y. The experimentally resolved FAST-PETase structure (PDB ID 7SH6; chain A; 1.44 Å resolution) served as the common structural template. Y87Q and S187Y were introduced on this scaffold, followed by local side-chain relaxation before substrate placement. The catalytic triad was retained as Ser160-His237-Asp206 in all three systems.

PET was represented by a trimeric oligomer, 2-hydroxyethyl-(mono-hydroxyethyl terephthalate)₃, referred to as the PET trimer. This model covers the central catalytic region and neighboring substrate-binding subsites while remaining tractable for long-timescale MD and quantum-chemical calculations, consistent with previous computational treatments of PET hydrolases [1–3]. Conformational sampling of the isolated oligomer was performed with CREST using the GFN2-xTB Hamiltonian [4,5], and low-energy structures representing distinct conformational families were retained as docking candidates.

For MD simulations, the electrostatic potential of the PET trimer was evaluated at the HF/6-31G(d) level and atomic partial charges were obtained using a two-stage RESP fit [6]. Bonded and Lennard-Jones parameters were assigned with GAFF2 through Antechamber in AmberTools [7], and the protein was described using ff19SB [8]. Hydrogen atoms were added

and titratable-residue protonation states were evaluated with PROPKA 3.0 at pH 8.0 [9]. The catalytic histidine was assigned with N δ protonated and N ϵ available for interaction with the catalytic serine. Crystallographic solvent and non-protein species not directly associated with substrate recognition were removed during receptor preparation.

Table S1. Structural templates and model construction for the FAST-PETase systems.

System	Template PDB	Resolution	Chain	Model construction
FAST-PETase	7SH6	1.44 Å	A	Direct FAST-PETase scaffold; no additional mutation introduced.
FAST-Y87Q	7SH6	1.44 Å	A	Y87Q introduced on the FAST-PETase scaffold, followed by local side-chain relaxation.
FAST-S187Y	7SH6	1.44 Å	A	S187Y introduced on the FAST-PETase scaffold, followed by local side-chain relaxation.

S3. Docking and molecular-dynamics simulations

Initial PET-binding configurations were generated with AutoDock Vina 1.2.0 [10,11]. Receptor and ligand PDBQT files were prepared with AutoDockTools in MGLTools 1.5.7. The protein was held rigid during docking, whereas chemically accessible PET torsions were allowed to rotate. Gasteiger charges were used only for Vina calculations and were replaced by RESP charges when selected complexes were converted to the Amber representation.

The docking search region was a $30 \times 30 \times 30$ Å box centered on the catalytic Ser160 O γ atom. Searches used an exhaustiveness value of 64 and retained up to 20 poses within a 5 kcal mol⁻¹ energy window. Docked poses were filtered using catalytic geometry rather than docking score alone: the scissile carbonyl oxygen was required to face the conserved oxyanion-hole donors and the electrophilic carbonyl carbon to remain within 4.5 Å of Ser160 O γ . Multiple geometrically acceptable solutions were retained for independent MD assessment when available.

Selected enzyme-PET complexes were immersed in cubic TIP3P water boxes extending at least 10 Å from the solute and neutralized with counterions. Simulations used the GPU-accelerated pmemd.cuda implementation in Amber 22 [12]. Solvent and ions were minimized first with restraints on enzyme and PET heavy atoms, followed by unrestrained whole-system minimization. Systems were heated from 0 to 300 K over 100 ps under Langevin temperature control, equilibrated under NVT conditions and then density-equilibrated in the NPT ensemble at 1 bar. A 2 fs integration time step was used during production.

S4. Replica design, PRS sampling and statistical analysis

Five independent MD simulations with independently generated initial velocities and random-number seeds were initiated for each enzyme-PET system [13]. The first 20 ns of each trajectory were treated as additional relaxation, and the common 20–100 ns interval was used for standardized comparisons across FAST-PETase, FAST-Y87Q and FAST-S187Y. Geometric PRS occupancies and other replica-level observables were evaluated independently over this common window and then summarized across the five replicas.

The geometric PRS for the acylation step was defined by a Ser160 O γ ...C_{acyl} distance of ≤ 3.5 Å together with an O γ -C=O Bürgi-Dunitz approach angle of 100–110°. Individual trajectory frames were not treated as independent statistical observations. Scalar observables were averaged within each replica and then combined across the five independent simulations; unless stated otherwise, values are reported as mean $\pm$ SD across replicas ($n = 5$).

A separate trajectory set was used for extended mechanistic characterization. Trajectories in which PET remained stably associated with the catalytic cleft and repeatedly sampled attack-compatible Ser160 O γ ...C_{acyl} distances during the initial production stage were extended to 1 μ s. In the current three-system dataset, all five replicas per system satisfied the active-site retention criterion. These extended trajectories were used for PRS-conditioned clustering, detailed noncovalent- and electronic-interaction analyses, QM-cluster calculations and DIAS, but not for estimating cross-system PRS populations. This separation allows PRS quantity to be evaluated from a common replica window while longer trajectories characterize catalytically relevant configurations after such states have been accessed.

Trajectory analyses were performed with CPPTRAJ [14]. Representative structures were selected by clustering equilibrated trajectory segments; for reaction-ready analyses, clustering

was repeated using only frames assigned to the geometric PRS ensemble. Molecular representations were generated with VMD 1.9.3 and PyMOL 3.1 [15,16].

S5. Quantum-chemical and electronic-structure analysis

All-QM cluster models were constructed from reaction-ready configurations sampled during MD. PRS frames were identified separately in each of the five replicas and clustered according to local active-site geometry. Four representative configurations were selected from populated PRS states in each replica, giving 20 PRS-derived QM models per enzyme. The same active-site composition and truncation strategy were applied to FAST-PETase, FAST-Y87Q and FAST-S187Y wherever structurally equivalent residues were available. Each cluster contained the catalytic Ser-His-Asp machinery, oxyanion-hole backbone N-H donors, the reactive portion of PET and neighboring residues required to preserve the principal local interaction network. Truncated boundaries were capped with hydrogen atoms.

Geometry optimizations were carried out with Gaussian 16, Revision A.03 [17], using the range-separated ω B97X-D functional with the 6-31+G(d) basis set [18]. Reactant-state, transition-state and tetrahedral-intermediate structures were optimized with SMD water as the continuum solvent [19]. Vibrational-frequency analyses at the same level were used to verify minima and first-order saddle points, with thermal corrections evaluated at 298.15 K and 1 atm. IRC calculations confirmed connectivity between optimized transition states and the corresponding reactant and tetrahedral-intermediate states [20].

The 20 calculated acylation barriers per enzyme were used to characterize conformational variability in the chemical step. Because four QM structures were sampled from each independent MD replica, the QM calculations were treated as nested conformational samples rather than as 20 independent statistical replicates. The four barriers from each replica were first averaged, and the five replica-level means were treated as independent statistical units; grand mean barriers were calculated across all 20 models.

Noncovalent and electronic interactions associated with catalytic preorganization were characterized using IGMH, QTAIM and NBO analyses. IGMH and QTAIM calculations were performed with Multiwfn 3.7 [21–24], and NBO analysis was performed with NBO 7.0.

S6. Distortion/interaction activation-strain analysis

DIAS analysis was used to decompose the energetic cost of reaching the acylation transition state into structural-deformation and interfragment-interaction contributions [25–27]. The QM cluster was partitioned consistently into enzyme and substrate fragments across FAST-PETase, FAST-Y87Q and FAST-S187Y. The distortion term describes the energetic penalty of deforming the individual fragments from their reactant-state geometries to the geometries adopted at the transition state, whereas the interaction term describes the interaction between the distorted fragments at the transition-state geometry. DIAS terms were interpreted as energetic descriptors of structural deformation and transition-state interaction, not as direct measures of the conformational free-energy cost of PRS formation

Supplementary Results

S7. Additional characterization of Y87Q

At both 30 and 60 °C, Y87Q showed backbone RMSD values comparable to those of FAST-PETase, while global hydration remained largely unchanged. The principal changes were localized near the active-site region. Active-site water occupancy decreased from approximately 6.5 to 3.6 waters at 30 °C, although this difference was less pronounced at 60 °C. These observations argue against global stabilization or generalized dehydration as the dominant origin of the enhanced phenotype.

The local second-shell interaction network showed a persistent directional contact between the substituted Gln side chain and Met161. The Gln carbonyl oxygen approached sulfur at approximately 3.0 Å with a C-S $\cdots$ O geometry consistent with a directional sulfur-centered interaction. NBO analysis identified lone-pair donation from the Gln carbonyl oxygen into the $\sigma^*(\text{C-S})$ orbital of Met161 with $E(2) = 2.9 \text{ kcal mol}^{-1}$. QTAIM identified a corresponding S $\cdots$ O bond path, and IGMH analysis supported a localized weak interaction. These analyses complement the ensemble-level PRS result.

Beyond the approximately twofold increase in geometric PRS population, extended trajectories supported persistent changes in active-site solvation and maintenance of a shorter substrate-catalytic-serine separation. Reaction-ready configurations more frequently coincided with oxyanion-hole engagement. Together, these observations support the interpretation that Y87Q

primarily shifts the bound ensemble toward configurations requiring less reorganization before nucleophilic attack.

S8. Additional characterization of S187Y and cross-scaffold validation

The nine individual point mutations generated using the structure-based design approach were experimentally evaluated. Among these mutants, S187Y proved to have the largest enhancement in activity in the FAST-PETase scaffold. Point mutations at the same position had very different activities, and some mutants, which had good docking scores, demonstrated lower activities compared to the parent enzyme. Therefore, the experimentally obtained activity ranks were not consistent with the predicted binding energy rankings, illustrating the discrepancy between the ground-state docked pose and productive catalysis.

The enhancement caused by the S187Y mutation was also noted for the ThermoStable-PETase scaffold, in which S187Y led to a 3.44-fold increase in the activity in degrading the PET film. This mutant showed its activity in a wider range of enzyme concentrations and was less inhibited by increasing enzyme concentration. Higher production of both MHET and TPA were found using UPLC analysis in both enzyme scaffolds. Overall, the catalytic role of the S187Y mutation can be concluded.

S9. Ensemble quantum-chemical analysis and mechanistic interpretation

Several QM structures were chosen from the PRS structure set to ensure that the results of the study would not be based on a single structure. Both advantageous mutants had different barrier distributions compared to the FAST-PETase. Furthermore, DIAS analysis indicated that the energy factors that caused such changes were different for each mutant. Thus, the most prominent factor for Y87Q was reduced distortional energy, while S187Y benefited from the interactions of the residues with the transition state.

The combination of these results with the results from the MD studies suggests two possible mechanisms for the enhancement of catalysis. While the Y87Q mutation seems to enhance the likelihood of attaining geometrically prepared states due to an increase in the population of PRS, S187Y mutation seems to enhance the efficiency of reactions of the prepared states due to more favorable transition state interactions. The mechanistic implications of the S187Y

mutation need to be seen with caution as the direct analysis of the PRS population and PRS conditioned catalytic parameters is required.

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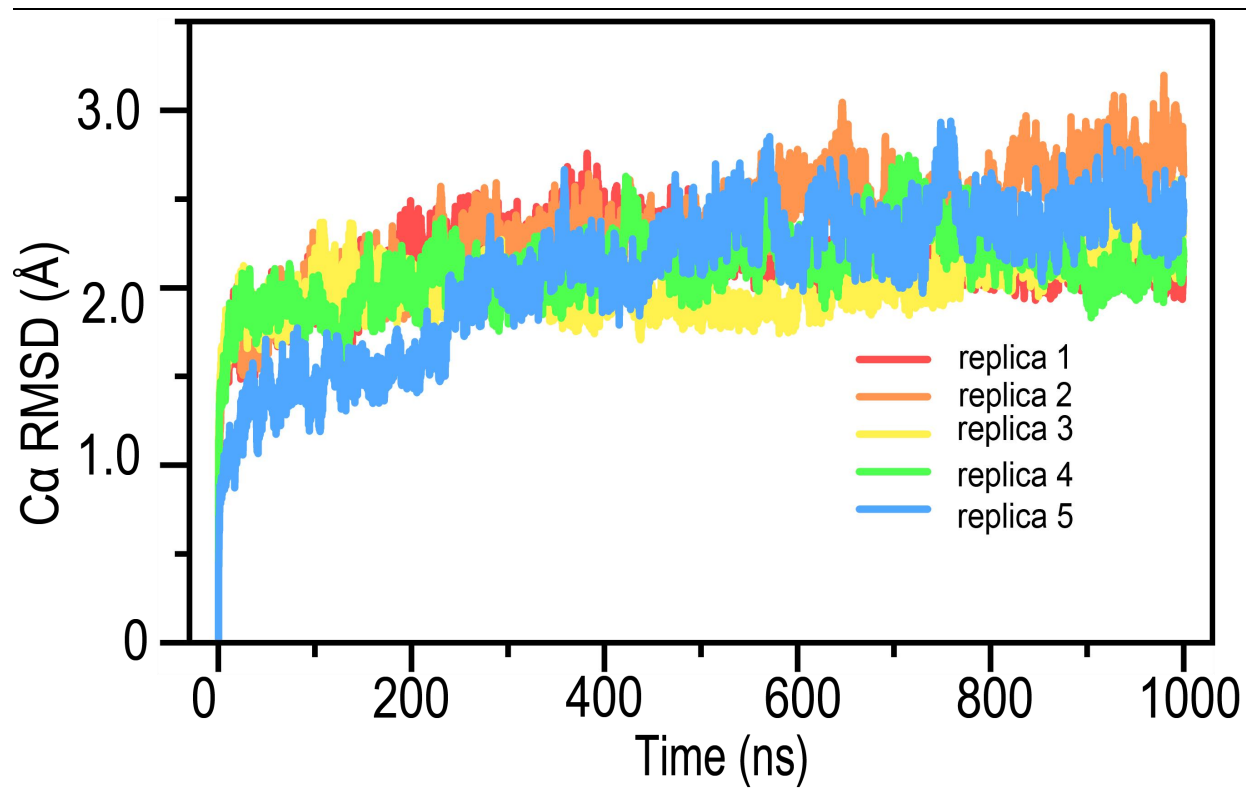


Figure S1. Global structural dynamics reproducibility in five separate MD replicates.